Table 3: OmniContext (Wu et al., 2026a), unseen during the search and scored by its GPT-4.1 evaluator. Cells are the geometric mean of prompt following and subject consistency (0–10) per task type, and Avg. the mean of the eight; Char.: character, Obj.: object, C.+O.: character and object. Best open image generators per column in bold, second best underlined. †: AutoRef-Harness discovered with FLUX.2 [klein] 4B on the four-reference MultiBanana setting.

Method	Gen.	SINGLE		MULTIPLE			SCENE			**Avg.**
		Char.	Obj.	Char.	Obj.	C.+O.	Char.	Obj.	C.+O.	
Proprietary Models										
GPT-Image-1.5	1	9.56	9.70	9.32	9.46	9.26	9.69	9.39	8.93	9.41
Nano Banana Pro	1	9.63	9.42	9.46	9.19	9.02	9.35	8.39	8.20	9.08
Seedream 4.5	1	9.45	9.50	9.09	9.39	9.09	9.35	8.66	8.23	9.09
Open Models										
OmniGen2	1	8.51	5.73	6.30	6.58	7.71	6.93	6.10	7.03	6.86
DreamOmni2	1	7.81	6.72	4.80	7.07	5.92	5.78	5.63	5.72	6.18
BAGEL	1	6.67	7.09	3.43	6.74	7.08	3.97	4.11	5.63	5.59
FLUX.2 [klein] 4B	1	**9.22**	8.16	7.91	8.21	8.68	9.18	7.47	7.54	8.30
+ **AutoRef-Harness**†	3	8.94	8.99	9.02	8.78	8.59	<u>9.28</u>	**8.89**	<u>8.28</u>	<u>8.85</u>
FLUX.2 [klein] 9B	1	<u>9.17</u>	9.08	8.66	8.14	<u>8.74</u>	**9.35**	8.07	7.47	8.59
+ **AutoRef-Harness**†	3	**9.22**	<u>9.18</u>	**9.19**	**9.32**	**8.82**	**9.35**	8.40	**8.30**	**8.97**
Qwen-Image-Edit-2511	1	9.14	**9.19**	8.66	<u>9.00</u>	8.33	6.97	8.17	8.17	8.45
+ **AutoRef-Harness**†	3	9.12	8.64	<u>9.09</u>	8.62	8.34	8.71	<u>8.82</u>	<u>8.28</u>	8.70

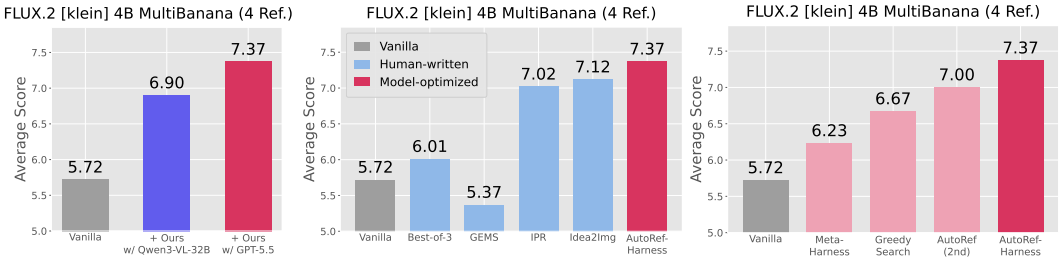


Figure 4: (**Left**) Reasoning-model transfer. AutoRef-Harness (+ Ours) improves FLUX.2 [klein] 4B with both GPT-5.5 and the open-weight Qwen3-VL-32B. (**Middle**) Comparison with human-written harnesses. AutoRef-Harness outperforms Best-of-3, GEMS, IPR, and Idea2Img. (**Right**) Comparison of harness optimization methods. AutoRef discovers stronger harnesses than Meta-Harness and Greedy Search; even its second-best discovered harness outperforms the final harnesses produced by both alternative search procedures.

and the harness structure unchanged, AutoRef-Harness improves FLUX.2 [klein] 4B from 5.72 to 6.90 on the four-reference MultiBanana held-out test split (7.37 with GPT-5.5; Figure 4, **Left**). This suggests that the orchestration strategy encoded by AutoRef-Harness is not specific to the proprietary reasoning model used during its discovery.

6.4 Comparison with Harnesses and Harness Search

Comparison with Human-Written Harnesses. We compare AutoRef-Harness with human-written harnesses: Best-of-3 (Ma et al., 2025), GEMS (He et al., 2026b), IPR (Oshima et al., 2026), and Idea2Img (Yang et al., 2024b), which use 3, 2.7, 3, and 9 image generations per task, respectively (details in Appendix G.1). AutoRef-Harness achieves the highest performance on the four-reference MultiBanana held-out test split (Figure 4, **Middle**). Notably, Idea2Img still underperforms despite using three times AutoRef-Harness's generation budget.

Comparison with Harness Optimization Methods. We next compare AutoRef with alternative harness optimization methods (details in Appendix G.2). Meta-Harness (Lee et al., 2026) uses the same tasks for optimization feedback and candidate ranking, and its proposer chooses which candidate to build on from the full search history. Greedy Search adopts AutoRef's train–validation separation but retains only the single best harness per iteration, whereas AutoRef keeps the top-B candidates